Independent audit of the Boussinesq fluctuation determinant foundations

Project audit

4 November 2026

Scope and result. This note audits the logical passage from the exact phonon solutions and their phase integrals to a physical one-loop two-soliton or soliton–defect amplitude. The good-equation and potential-field conventions of the repository are retained. No regularisation, asymptotic state convention or exchange phase is selected. The nonconstant continuum integral has substantial positive evidence. Its identification with the full determinant, including zero-mode measure and an absolute phase, remains incomplete **(d)**. The label **(d)** here denotes an unestablished physical conclusion, not a counterexample to the continuum integral. All new numerical and symbolic identities below are checked by `scripts/audit_determinant_foundations.py`, at 65-digit working precision; raw results are saved in `scripts/audit_determinant_results.json`.

1 What the cited determinant argument supplies

Faddeev–Korepin (FK) [1] derives the relevant one-loop determinant ratio for relativistic scalar fields, with the Klein–Gordon fluctuation operator. In §5.4, printed p. 62, the constant factor Z is explicitly left to supplementary requirements, with crossing symmetry as an example. Equation (4.28), printed p. 66, still contains Z. This is a statement of the source's scope **(a)**, not a Boussinesq theorem. The source's Appendix 6 derives its spectral density from a second-order Schrödinger equation, and Appendix 5, p. 81, uses localization of discrete modes to separate their Wronskians.

The Boussinesq operator has fourth-order spatial derivatives and derivative coupling to the background. A canonical second-order time derivative ensures the form of the temporal symplectic product; it does not alone establish completeness, the resolvent, the infrared domain, or the source's entire spectral-density derivation. Thus "apply verbatim" in `oneloop_phase.tex` is too strong. The three-soliton tau linearisation provides an exact continuum family with diagonal transmission. Conditional on its completeness in the relevant scattering space, this supports a vanishing continuum mixing kernel. It does not itself construct the complete projected resolvent with collective-coordinate directions removed **(d)**.

A co-moving sign correction. The displayed co-moving ODE in Phase 2e has a typographical sign error. Direct symbolic differentiation gives **(c)**

$$\partial_t^2\{e^{-i\widetilde{\omega}t}f(x-vt)\} = e^{-i\widetilde{\omega}t}(-\widetilde{\omega}^2 f + 2iv\widetilde{\omega}f' + v^2 f'').$$

Its first-derivative term therefore has a plus sign. The subsequent quartic in that report uses the correct sign; the lab-frame PDE checks are unaffected.

Positive control: the potential-field density. A direct calculation does support the one-soliton density step **(c)**. Write $k = \sin\alpha$, $v = \epsilon\cos\alpha$, $\omega = |q|\sqrt{1+q^2}$, $h = (1+\tanh(k\xi/2))/2$,

$A = e^{i\theta(q)}$ and

$$g(\xi) = 1 + (A-1)\left(h + \frac{h'}{iq}\right), \qquad \psi(x,t) = \omega^{-1/2} g(x-vt) e^{iqx - i\omega t}.$$

This is the potential-field tau mode, normalized to temporal plane-wave Wronskian density $-2i$. Direct spatial quadrature gives

$$\frac{1}{\omega}\int_{\mathbb{R}} \left[-2i\omega(|g|^2 - 1) - v(\bar{g}g' - g\bar{g}')\right] d\xi = -2i\,\theta'(q). \tag{c}$$

Four checks include both soliton orientations and both phonon directions. This checks the normalization that is implicit in the quoted $-4\pi i$ continuum Wronskian. It is not a check of the full time-dependent determinant.

2 Endpoint constants and exchange

For the phase continuous on each half-line and zero at both infinities, the light-branch endpoints are **(c)**

$$\theta(0^{\pm}; \alpha, \epsilon) = \pi(\pm 1 + \epsilon), \qquad 0 < \alpha < \pi/3. \tag{c}$$

Define the ordinary half-line integrals (without a delta function at zero)

$$I_{12} = \frac{1}{4\pi}\int_{\mathbb{R}\setminus\{0\}} \theta_1 \theta_2'\, dq, \qquad J_{12} = \frac{I_{12} - I_{21}}{2}.$$

Their integration-by-parts identity, for all four direction combinations, is

$$I_{12} + I_{21} = -\frac{\pi}{2}(\epsilon_1 + \epsilon_2), \tag{c}$$

$$J_{12} - I_{12} = \frac{\pi}{4}(\epsilon_1 + \epsilon_2), \tag{c}$$

$$I(\alpha, \epsilon; \alpha, \epsilon) = -\epsilon\frac{\pi}{2}. \tag{c}$$

The `oneloop_phase.tex` same-direction expression $-\pi$ is correct for $++$ and has the wrong sign for $--$. In particular, the later claims that raw I is antisymmetric and vanishes on the diagonal are false. The displayed identities are checked by direct quadrature of a phase reconstructed from the KP transmission ratio, independently of the closed dilogarithm formula.

The endpoint constants are independent of the amplitudes inside each regular chamber. Consequently the FK gradient cannot distinguish I from J or fix a further chamber constant. Phase 2e itself acknowledges the Z ambiguity and records the PI's decision to display both functions. A later preference for one function because it closes a KP identity is a normalization criterion, not a calculation of that determinant constant. In particular, a phase integral evaluated after both particle labels are swapped is not automatically the inverse physical collision: the ordering of incoming trajectories, the phase of the asymptotic state basis, and any exchange convention must also be specified. Neither the raw exchange failure nor algebraic closure alone settles physical unitarity **(d)**.

For the defect integral, setting the defect phase to zero at $q = 0^{\pm}$ makes the integration-by-parts term vanish. That is a precise statement about a chosen branch. The absence of a translation mode does not select a logarithm branch: $z_{\mathrm{ph}}(0) = 1$ fixes it only modulo 2π. No new defect branch is chosen in this audit. A physical fusion rule additionally requires normalized intermediate defect states and fusion amplitudes; continuum chord identities alone do not determine those data **(d)**.

3 Collective coordinates and the Maslov assertion

At $t = 0$, choose the repository potential gauge

$$\phi(x; \alpha, X) = k\{1 + \tanh[k(x - X)/2]\}, \qquad \pi = -v\phi_x.$$

The single-soliton symplectic pairing verifies the canonical momentum **(c)**:

$$\Omega(\partial_\alpha, \partial_X) = \int (\pi_\alpha \phi_X - \pi_X \phi_\alpha) dx = \frac{dP}{d\alpha}, \qquad P = \frac{2}{3}\epsilon \sin^3\alpha \cos\alpha. \tag{c}$$

However the amplitude variation is not a spatially localized potential mode:

$$\phi_\alpha(-\infty) = 0, \qquad \phi_\alpha(+\infty) = 2\cos\alpha. \tag{c}$$

The variation is a generalized collective-coordinate direction, including a secular component at nonzero time, rather than a second ordinary oscillatory bound state. Statements about localized δu cannot be transferred to the potential-field determinant without tracking this tail.

For two widely separated asymptotic profiles, soliton 1 on the left and soliton 2 on the right, retaining these tails gives **(c)**

$$\Omega(\partial_{\alpha_1}, \partial_{\alpha_2}) \longrightarrow -M_1' \Pi_2', \qquad M = 2\sin\alpha, \quad \Pi = -2\epsilon \sin\alpha \cos\alpha. \tag{c}$$

The check evaluates the spatial integral at separations 200 and 400. This is an asymptotic-profile calculation, not a substitute for the exact collision modes. It establishes that separation alone does not make the mixed block vanish. Darboux coordinates may remove these cross terms, but the corresponding zero-mode basis, normalization and measure must actually be supplied. Conservation of a Wronskian establishes time independence; it does not by itself establish equality to a product of one-body factors.

What the large-time Hessian does show. The checked derivative is **(c)**

$$\frac{dv}{dP} = -\frac{3}{2\sin 3\alpha} < 0, \qquad 0 < \alpha < \pi/3. \tag{c}$$

For fixed parameters away from degeneracies and resonance lines, a real symmetric finite correction to $T\,\mathrm{diag}(dv_j/dP_j)$ cannot change its negative signature at sufficiently large T. A matrix control in the script keeps the off-diagonal terms; the diagonal entries quoted as "eigenvalues" in Phase 2e are not generally eigenvalues. The valid signature observation concerns the endpoint stationary-phase wave-packet factor. It is not a calculation of spectral flow of the field fluctuation operator or of the branch of its square root. It therefore does not establish a full relative Maslov index or determine Z **(d)**. The cited Phase 2e script contains no Wronskian, measure, Hessian or Maslov test despite being cited for that claim.

A triangular coordinate shift with unit finite-dimensional Jacobian and a classical function of commuting momenta do rule out a particular elementary ordering problem. They are not an evaluation of the projected functional measure or its global phase. Thus the claimed absence of every additional one-loop topological/measure contribution is unproved **(d)**.

4 Localized defect modes and the physical domain

Phase 2g asserts that localized defect modes cancel because the defect has identical endpoint frequencies before and after a soliton crosses. Neither of its cited scripts evolves the localized modes through the collision. Endpoint frequencies alone do not prove cancellation. As a logical

control, an oscillator with frequency ω before and after a finite interval of length T and frequency W inside has the exact negative-frequency coefficient **(c)**

$$\beta = i\frac{W^2 - \omega^2}{2\omega W}\sin(WT). \tag{c}$$

At $(\omega, W, T) = (1, 2, 1)$ it is nonzero. The script derives this coefficient from the exact transfer matrix and verifies the symplectic normalization. This counterexample refutes the general implication used in the text; it does *not* assert that this mixing occurs in the integrable defect. That would require its actual localized scattering matrix. In the strictly stable regime with no localized defect modes this particular obstruction is absent; at thresholds and unstable vacua it cannot be dismissed by the same endpoint argument **(d)**.

The most defensible domain of a putative elastic two-soliton interpretation is the light branch with a nonsingular real two-soliton background, distinct velocities, and parameters away from cusp/resonance thresholds. Continuum integrals can remain finite beyond this domain without defining asymptotic stable scattering states. The tau construction supplies candidate discrete modes and the continuum phase winding, but a complete spectral theorem for the fourth-order operator pencil (with its zero-frequency threshold and potential-field domain) is not demonstrated by the existing finite samples. A "Levinson-type count" should remain distinguished from such a theorem **(d)**. The defect stability formula itself is audited separately.

5 Claims and checks

Claim	Label	Evidence
Normalized potential-mode density identity	(c)	Four 65-digit spatial quadratures
Orientation-dependent endpoint and exchange constants	(c)	Four channel quadratures, diagonal checks
Single and mixed asymptotic symplectic pairings	(c)	Direct field integrals and symbolic momentum derivative
Endpoint Hessian signature at large time	(c), limited	Full matrix control; no field spectral-flow claim
Equal localized frequencies need not imply cancellation	(c)	Exact oscillator transfer matrix
Full Boussinesq determinant and collective measure	(d)	Missing complete projected resolvent and normalization
Absolute one-loop phase, exchange/fusion constants	(d)	FK source retains Z; no independent anchor

6 Failures recorded

1. The co-moving ODE has the wrong first-derivative sign; its subsequent quartic and lab-frame PDE checks have the correct sign.

2. Prior orientation-independent same-direction boundary formula fails for $--$: its exchange sum is $+\pi$.

3. Prior raw antisymmetry and zero diagonal claims fail; the diagonal is $-\epsilon\pi/2$.

4. Localization-only cancellation of amplitude-mode Wronskians fails for the potential-field variations: their mixed asymptotic pairing is nonzero. This does not disprove cancellation after a correctly constructed Darboux measure; that calculation is missing.

5. Equal asymptotic localized frequencies do not imply determinant cancellation, as the exact pulse control demonstrates.

6. The first Hessian illustration incorrectly required an eigenvalue to differ from a diagonal entry by more than 1; this arbitrary bound failed. It was replaced by the exact symbolic characteristic-polynomial identity at each diagonal entry, which equals minus the square of the off-diagonal entry and is therefore nonzero in the test. No tolerance was weakened. The checks do not evaluate a functional determinant or prove completeness and are not reported as doing so.

References

[1] L. D. Faddeev and V. E. Korepin, *Quantum theory of solitons*, Physics Reports **42** (1978), 1–87, DOI:10.1016/0370-1573(78)90058-3. The publisher's entry and DOI were verified. Relevant full-text passages in §5.4 and Appendices 3–6 were inspected from the local PDF `Faddeev and Korepin - 1978 - Quantum theory of solitons.pdf`; this audit does not claim to have read the entire article.